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Mini project 2

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10. Mini project 2

Multiple Object Tracking (MOT) is a fundamental task in perception systems based on computer vision. It makes it possible to obtain information about the trajectories of objects present in a scene, which enables collision avoidance or the tracking of a specific target – Figure 10.1. It is used, among others, in autonomous vehicles, advanced video surveillance, and targeting systems.

Since it is not known in advance how many objects will be tracked or when they will appear, the object tracking task requires some detection system. The task then boils down to deciding which of the previously observed objects corresponds to each obtained detection. However, the tracking system should consider a number of scenarios that may occur in such a defined task, including:

- A detection may not correspond to any of the known objects because a new object has just appeared in the scene.
- A detection may be an error of the detector (*false positive*).
- An object may still be present in the scene and visible, but not detected by the detector in a given frame (*false negative*).
- One object may be occluded by another, and a single detection may overlap with the positions of two objects. The tracking system may then confuse the identities of the two objects (*identity switch*).

10.1 Project aim

The aim of the project is to develop a multiple object tracking algorithm based on a predefined set of detections. This makes it possible to focus on the tracking task itself and to decouple the influence of the detector on the quality of the entire system during performance evaluation. One of the most popular methods for evaluating tracking quality is the MOTA (Multiple Object Tracking Accuracy) metric.

$$MOTA = 1 - \frac{\sum_t (FN_t + FP_t + IDSW_t)}{\sum_t g_t} \quad (10.1)$$

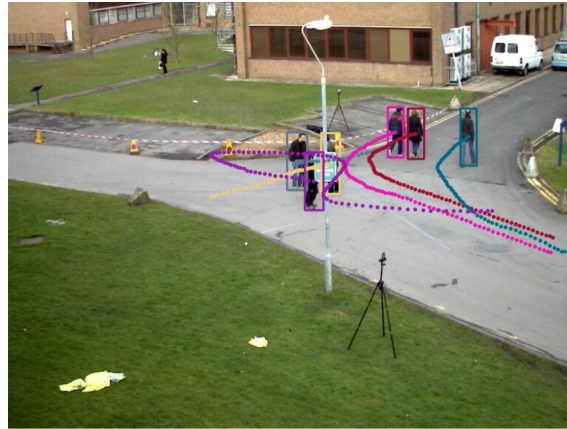


Figure 10.1: Multiple Object Tracking objective.

where t are the frames of the test sequence, FN_t is the number of *false negatives*, FP_t is the number of *false positives*, $IDSW_t$ is the number of *identity switches*, and g_t is the actual number of objects in frame t .

As part of the project, you must:

- interpret and visualise the dataset.
- propose and implement a multi-object tracking algorithm. You should consider the fundamental tasks: initialising a new object, assigning new detections to existing objects (based on position and appearance), and removing objects.
- carry out an evaluation of the tracking performance.

The dataset is available at: [link](#). For evaluation, it is worth using the official tool employed in all competitions related to multiple object tracking: [TrackEval – GitHub](#). The provided dataset is already in a compatible format.

10.2 Example literature

- Bewley, A., et al.: Simple online and realtime tracking.
- Zhang, Y., et al.: Bytetrack: Multi-object tracking by associating every detection box.
- Aharon, N., et al.: Bot-sort: Robust associations multi-pedestrian tracking.
- Bernardin, K., Stiefelhagen, R.: Evaluating multiple object tracking performance: The clear mot metrics.